Batch mode: define a dataset, process it, read it back

Developer quick-sheet · goldsluice and goldpan share one dataset file · version 1, 27 October 2026

A **dataset definition** is one small text file that says *which raw files*, *how to read them* and *which processing to run*. `goldsluice` runs it over every raw file in one command and writes the results next to it; `goldpan` opens the same file and gives you the pulses together with the processed quantities. Nothing is assumed: the readout, the channels and every parameter are written in the file. This sheet is the whole loop; the file format is specified in the docstring of `python/goldpan/dataset.py`, and a new algorithm is a different sheet (*Running goldsluice and developing an algorithm*).

You need the setup of that sheet (`maturin develop --release` in `goldsluice/`) and `goldpan` installed with Parquet support (`pip install -e "./python[parquet]"` from the repository).

1. The definition file

A TOML file, by convention `<name>.dataset.toml`. Annotated; the whole file is `examples/r76.dataset.toml`:

```toml
[dataset]
name = "r76-07221203"
output = "~/goldsluice-output/r76" # the processed quantities: outside the repository

[[sources]]                        # where the pulses come from; several allowed
name = "side2"
format = "idx"                     # any goldpan adapter: idx, midas, parquet, ...
glob = "~/idx/07221203_2025_F*"    # raw files or folders: paths = [..] and/or glob
detector = "NAI-S2"
readout = "DCRC_RUN76"             # a goldpan.config preset, or three numbers
[sources.read]                     # the adapter's other read() arguments
channels = { Side2pC = [0, 0], Side2pF = [0, 1] }
roles = { Side2pC = "phonon", Side2pF = "phonon" }

[[processing]]                     # the steps, in order
algorithm = "peak"
channels = "role:phonon"           # "all", "role:<role>", or a list of names
[processing.params]
glitch_samples = "@readout.glitch_samples"       # from the readout above
pretrigger_samples = "@readout.pretrigger_samples"
amps_per_adc = 1.5e-9              # you must supply it: there is no default

[[processing]]                     # the next step, run in order
algorithm = "max_time"             # time of the largest sample, in s
channels = "role:phonon"
[processing.params]
sample_period_s = "@readout.sample_period_s"   # from the readout
```

Key	Rule
`readout`	A preset name or `{ pretrigger_samples = ..., glitch_samples = ..., sample_period_s = ... }`, all three. Only needed by adapters that ask for it (`idx`, `midas`); a `parquet` source has its readout inside the file.
`[sources.read]`	Exactly the keyword arguments of that adapter's `read` (`help(goldpan.adapters.idx.read)`); an argument it does not have is an error.
`"@readout.<field>"`	In a parameter: the value of `pretrigger_samples`, `glitch_samples` or `sample_period_s` of the data's readout, so the file cannot disagree with itself.

Unknown keys	**Errors**, not ignored: a typo (`readuot`) fails at once instead of silently doing nothing.

2. Check, then run

```
python -m goldsluice algorithms
python -m goldsluice check r76.dataset.toml
python -m goldsluice run r76.dataset.toml
```

`algorithms` lists what can be named in `algorithm =` and the outputs of each. `check` reads the definition and lists the raw files it finds; it fails for an unknown algorithm, a pattern that matches nothing or a bad readout, before anything is processed. `run` processes the files and ends with a report that times each phase:

```
processed  .../raw/TOY/S1_F0001.parquet: 3000 events in 0.75 s
processed  .../raw/TOY/S1_F0002.parquet: 3000 events in 0.65 s
dataset big: 2 file(s) processed, 0 unchanged, 6000 events in 1.49 s
  read        1.244 s   (the adapters)
  compute     0.052 s   (peak)
  write       0.101 s
       4021 events/s overall
```

That is 6000 events of 3 channels and 4096 samples from a Parquet source. **Reading is usually most of the time**; the Rust kernels the least. Options: `--chunk N` (events read at once, default 1000: it bounds the memory used by the pulses), `--force` (redo files that are unchanged) and `--backend python` (the Python mirrors: slow, to compare numbers).

Results do not belong in the repository. `check` and `run` refuse an `output` folder that lies inside a Git working tree whose `.gitignore` does not cover the data files written there, and say so before anything is written. The repository ignores `processed/` and `_goldsluice.json`, and the example writes to `~/goldsluice-output/`, so the common cases pass. If you really mean it, add `--allow-output-in-repo`.

One output root. Set the environment variable `GOLDSLUICE_OUTPUT` to a folder (for example `C:/NSDF/goldsluice-output`) and a *relative* `output` in a definition (`output = "r76-nai"`) goes under it, so the file holds no path of your machine and runs unchanged on another computer. A full path (`output = "~/goldsluice-output/r76"`) is used as written; with the variable unset a relative `output` is relative to the definition file, as before. The goldpan writers (`parquet.write`, `rq_root.write`) write exactly where they are told and do not check the folder: the guard is batch mode's.

Steps that build on each other. A step reads the raw trace unless it has an `input`. An algorithm that gives a *trace* instead of numbers (a filter or a baseline subtraction: `python -m goldsluice algorithms` says which: `lowpass`, `subtract_line`) needs a `name` and writes no quantity; a later step with `input = "<name>"` reads its result instead of the raw trace. A `suffix` is added to the names of the quantities of a step, so the same algorithm can run on both (`adc_max` and `adc_max_lp`). An input is a step *above* the one that reads it, and its channels must include the channels of the reader. The example `examples/r76-registry.dataset.toml` has two chains:

```toml
[[processing]]
algorithm = "lowpass"              # gives a trace: it has a name
name = "lp"
channels = "role:phonon"
[processing.params]
cutoff_hz = 95e3
order = 4
sample_period_s = "@readout.sample_period_s"

[[processing]]
algorithm = "peak"                 # reads the filtered trace
channels = "role:phonon"
input = "lp"
suffix = "_lp"                     # adc_max_lp, current_max_lp
[processing.params]
# glitch_samples, pretrigger_samples, amps_per_adc ...
```

Several sources, and different steps for each. A definition can have several `[[sources]]`, each with its own files,

format and `detector` (from the registry), so raw data from detectors that are wired differently go through one definition. Every step runs on every source, and `channels = "role:phonon"` means each detector's own phonon channels. When the detectors need different processing, a step names the sources it runs on: `sources = ["ge"]`. A step without `sources` runs on all of them. A step that reads a filter (`input`) must run on a subset of the filter's sources, and every source needs at least one step. A source's files are redone only when the steps that run on it change. The sources' different quantities read back together as records with different fields, so pick a source's rows by `series` first. `output_format = "root"` is for one detector only.

Run metadata. Facts about whole series (high voltage, trigger setting) that an analysis needs for every event go in a small CSV, one row per series, with a `series` column and any others (`hv_v`, `trigger_value`, ...; see `examples/r76.series.csv`). The definition only references it: `metadata = "r76.series.csv"` under `[dataset]` (relative to the definition file; later it may name a database instead). The values are **joined when the data are read**: `Dataset.processed()` adds one column per column of the table to every event, so a change of a value never redoes a file, and `goldsluice run` does not write them. A series with events that is not in the table is an error that names it; so is a column named like a quantity or a channel; `processed(metadata=False)` leaves them out. For events you already have, `goldpan.runmeta.per_event(series, table)` gives one value per event and `groups(series, table, by=("hv_v", "trigger_value"))` the events of each chunk.

3. Read it back with goldpan

```
from goldpan.dataset import Dataset

ds = Dataset.open("r76.dataset.toml")
ev = ds.read()          # the pulses with the processed quantities: one Events
q = ds.processed()      # only the processed quantities, no pulses: small, fast
ev.quantities["Side2pC"]["current_max"]    # a record per channel, a field per output
```

`ds.iter_events()` gives one `Events` per raw file (for a dataset larger than memory); `ds.read(processed=False)` gives only the pulses. Both packages read the *same* file, so a notebook and a batch job cannot disagree about what the dataset is. Reading before the batch run raises an error that names the command to run.

4. What is written, and what makes it run again

```
~/goldsluice-output/r76/
    _goldpan.json        the goldpan Parquet manifest
    _goldsluice.json     the ledger: per raw file, what was done, with what, how long
    NAI-S2/07221203_2025_F0001.parquet    one file per series and dump:
                         the key (series, dump, event) and the quantities
```

Any other tool that reads Parquet (pandas, Polars, DuckDB) opens these files. Each carries its provenance (the steps and parameters, the goldsluice version, the timing) in its metadata under `goldsluice`. A raw file is **processed again** when its size or date changes, or when anything that determines its results changes: the way it is read, the readout, a step or a parameter. Renaming the dataset or moving the output does not count. To start over, delete the output folder or run with `--force`.

5. Adding steps

Add another `[[processing]]` table. Each quantity must come from **one** step (two steps giving `Side2pC.adc_max` is an error that names it); different algorithms add different fields to the same channel's record. Use `channels = ["Side2pC"]` to process only some channels. For an algorithm that does not exist yet, write its Python mirror and Rust kernel (the other sheet): `algorithms` lists it as soon as it is registered.

6. ROOT files instead of Parquet

Two lines in `[dataset]` write the quantities as ROOT files laid out like the collaboration's RQ files (they need `uproot`: `pip install goldsluice[root]`):

```
[dataset]
name = "r76-07221203"
output = "~/goldsluice-output/r76-root"
```

```
output_format = "root"              # "parquet" is the default
root_detector = 1                   # N of the zip<N> tree: required, never assumed
```

There is one file per series and dump, `<series>_<dump>.root`, with `rqDir/eventTree` (`SeriesNumber`, `DumpNumber`, `EventNumber`) and `rqDir/zip<N>` row-aligned, every branch a `double`, as in the real RQ files. The branches keep **our names**, `Side2pC_adc_max` and not the collaboration's `PAS1max`: ROOT is only the container. The provenance of the run is stored as the string `infoDir/goldsluice`. The ledger, the chunks and the output-location guard work as for Parquet.

Read the files back with the RQ reader of goldpan (`Dataset.processed()` reads Parquet output only, and says so):

```
from pathlib import Path
from goldpan.adapters import rq_root

t = rq_root.read(Path.home() / "goldsluice-output" / "r76-root", detector=1)
t["Side2pC_adc_max"]        # one record per event
```

`root_detector = 1` is **our choice** for the NaI detector of Run 76: the number is not in the raw data, so it lives in the definition file; use the same one in every definition of that detector. (The collaboration's Z7 files have `zip7` and `zip14`.) The commented lines in `examples/r76.dataset.toml` show the same two keys.

7. When it fails

Message	Meaning and fix
`unknown key(s) [...]`	A typo or a key that does not exist; the message lists the allowed ones.
`...matches nothing`	A `paths` or `glob` entry found no file; the message says where it looked (relative paths start at the definition file).
`readout needs [...]`	An inline readout lacks numbers; there are no defaults.
`needs a readout`	The adapter (`idx`, `midas`) takes a readout and the source gives none.
`has no argument [...]`	`[sources.read]` names something the adapter's `read` does not take.
`unknown algorithm`	Not in `algorithms`; check the spelling or register the algorithm.
`missing ... keyword-only argument`	A step lacks a parameter: algorithms have no defaults.
`nothing processed yet`	You read the dataset before `run`.
`has the batch run covered this file`	New raw files were added after the last run; run again.
`inside the Git working tree`	The `output` folder is in a repository that does not ignore it: write it elsewhere (`~/goldsluice-output/...`), ignore it, or pass `--allow-output-in-repo`.
`needs root_detector`	`output_format = "root"` without the number of the `zip<N>` tree.
`rq_root.read` (from `processed()`)	The output is ROOT: read it with `rq_root.read`, not `Dataset.processed()`.

The numbers in `current_max` are in **ADC counts times** `amps_per_adc`: a placeholder of `1.0`, as in the example file, gives counts.